

Istiaqur Rahman Fuad

📍 Jamalpur | ✉ istiaqurrahmanfuad@gmail.com | ☎ +880 1865-863271

🌐 istiaqfuad | 🔄 istiaqfuad | 🌐 istiaqfuad.me

SUMMARY

Computer science undergraduate and aspiring backend software engineer specializing in scalable systems and APIs. Comfortable architecting end-to-end web applications and integrating modern AI features as secondary capabilities. Enjoys building robust infrastructure, optimizing databases, and open-sourcing the work.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, C++, C, SQL

Technologies & Tools: Spring Boot, Next.js, FastAPI, PostgreSQL, Redis, RabbitMQ, Docker, Prisma, Stripe, Git

AI / ML: PyTorch, Hugging Face, LoRA, LLMs, LangChain, LangSmith, Vector Databases

EXPERIENCE

Advanced Machine Intelligence Research (AMIR) Lab

Feb 2026 – Present

Research Assistant Intern

Remote

- Curated and published **bangla-english-banglish-pairs**, a 2.4M-pair trilingual contrastive dataset (Bangla / Banglish [romanized Bengali] / English) on Hugging Face for low-resource NLP research.
- Trained and open-sourced **TriBne-e5-small**, a trilingual sentence-embedding model for retrieval over code-mixed Bengali text.

Devcore Technologies

Feb 2025 – Apr 2025

Full Stack Developer Intern

Remote

- Architected a modular admin dashboard and end-to-end type-safe backend APIs using tRPC and Prisma for an e-commerce platform with a 1,000+ product catalog.
- Developed responsive, server-rendered storefront pages with Next.js App Router, improving cross-device usability.

EDUCATION

University of Rajshahi

Jan 2022 – Expected 2026

B.Sc. in Computer Science and Engineering

CGPA: 3.58/4.00

- Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Software Engineering, Artificial Intelligence, Machine Learning

PROJECTS

EventHub | [Live](#) | [GitHub](#)

- Tools used:** Java, Spring Boot, PostgreSQL, Redis, RabbitMQ, Next.js, Docker, Stripe
- Engineered a scalable full-stack ticketing platform (Spring Boot, Next.js) containerized with Docker Compose.
- Implemented interactive visual seat selection with Redis holds, Stripe Checkout, and RabbitMQ for async ticket processing.

LegalBuddy | [Live](#) | [GitHub](#)

- Tools used:** Python, FastAPI, Next.js, Qdrant, sentence-transformers, LLM, Docker
- Built a retrieval-augmented generation (RAG) system over Bangladesh legal acts: a FastAPI backend retrieves grounded statute sections from Qdrant and generates cited answers with LLM in a Next.js chat UI.

RESEARCH & PUBLICATIONS

A Trilingual Contrastive Corpus and Parameter-Efficient Recipe for Romanized Bengali Retrieval

- First author;** under review at *Language Resources and Evaluation* (Springer Nature), 2026. [Demo](#) | [GitHub](#)
- LoRA fine-tuned multilingual-e5-small (1% parameter update, MNRL loss) on a self-built 2.4M-pair trilingual (Bangla / Banglish / English) contrastive corpus for Banglish and cross-script retrieval.
- Reached state-of-the-art on Banglish-spelling and cross-script retrieval, lifting cross-script MRR@10 from 0.50 to 0.98 over the base model and beating BGE-M3, LaBSE, and Embedding Gemma.

COURSES AND CERTIFICATIONS

- Machine Learning Specialization – Stanford University | DeepLearning.AI (Jul 2023). [certificate](#)
- Deep Learning Specialization – Stanford University | DeepLearning.AI (Oct 2023). [certificate](#)

ACHIEVEMENTS AND ACTIVITIES

- Award of Merit** – International Blockchain Olympiad (Oct 2024). [certificate](#)
- SDG-9 Silver Medal** – International Blockchain Olympiad (Oct 2022). [certificate](#)